Significance Sorting in Government Intervention Research: Evidence from Top Finance Journals*

[Author Name]† [Co-Author Name]‡

This version: May 29, 2026

[Working paper — please do not cite without permission]

ABSTRACT

We document a significance-sorting pattern in the three leading finance journals—the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics*—using a novel sample of government intervention research spanning 2000–2024. Combining a keyword filter with a large language model (LLM) classifier, we identify 296 in-scope papers and code the direction of their primary empirical finding. Our primary result: papers with *any* directional finding are published in top journals at a rate of 83–85%, compared with only 41% for null-finding papers—a gap of 43–44 percentage points that is robust across probit specifications ($p < 0.001$). This finding does not depend on how we classify the sign of the finding: a binary directional indicator (directional vs. null) yields a coefficient of 1.241*** (s.e. 0.162), sidestepping normative direction coding entirely. Within directional papers, we find no evidence of directional asymmetry: positive-finding and negative-finding papers are published at virtually identical rates (83.6% and 85.6%, $p = 0.73$). The pattern is concentrated in the *Review of Financial Studies* and the *Journal of Financial Economics* and

*We thank seminar participants at [XX] for helpful comments. We are grateful to [XX] for outstanding research assistance. All data and code are available at https://github.com/[REPOSITORY].
Correspondence: [author@university.edu].

†[Department of Finance, University of XX.]

‡[Department of Economics, University of YY.]

is present across all sub-periods. We interpret these results as a cross-sectional association and are explicit that our design cannot distinguish editorial selection from author self-selection in submission.

JEL classification: G18, G28, C80, B40.

INTRODUCTION

Academic research on government intervention in financial markets occupies a privileged position in public policy. The Federal Reserve Board, the Securities and Exchange Commission, and the Basel Committee on Banking Supervision routinely incorporate peer-reviewed finance research into regulatory impact assessments, rule-makings, and policy speeches. The Dodd-Frank Wall Street Reform and Consumer Protection Act alone generated an estimated 400 new regulatory mandates, many of which were designed in explicit reference to academic evidence on the effects of capital requirements, disclosure rules, and consumer financial protection. The credibility of this evidence base is therefore a first-order policy question: if the academic record systematically misrepresents which interventions have measurable effects—and which do not—regulators face a distorted picture that can generate welfare-reducing policy choices.

Publication bias offers a mechanism by which such distortion arises without any deliberate misconduct. When researchers are reluctant to write up null results, when journals are more likely to accept papers with statistically significant findings, or when both effects operate simultaneously, the published literature understates the true prevalence of null or inconclusive evidence (Franco et al., 2014). In a domain such as government intervention research, where policy decisions depend on whether a regulation has a measurable causal effect, suppression of null results generates the false impression that most interventions work—or, more precisely, that most interventions have discernible effects in one direction or the other.

This paper provides the first systematic evidence on publication bias in the finance literature's treatment of government intervention. Our research design combines two complementary samples. The *published paper* sample consists of all in-scope articles published in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024. The *working paper* sample is drawn from NBER working papers in five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE), covering papers from 2000 to 2024. Together, these samples allow us to compare the distribution of findings across publication states in this domain—not merely the subset that appears in top journals.

We define a paper as in-scope if its primary research question is the causal effect of a specific government law, regulation, or public policy on financial markets, firms, or investors. Identifying in-scope papers in a corpus of more than 33,000 documents would be infeasible by hand. We therefore develop a two-pass automated pipeline: a keyword filter that screens for government intervention terminology, followed by a large language model (LLM) classifier that reads each flagged abstract and returns a binary in-scope decision with a structured justification. For each in-scope paper we then code the *direction* of the primary finding as positive (+1, the intervention was beneficial), negative (−1, harmful), or null/mixed (0).

Our primary finding is a pronounced *significance-sorting* pattern in the published record: papers with *any* directional finding appear in top finance journals at substantially higher rates than null-finding papers. A binary directional indicator (directional vs. null, sidestepping normative positive/negative coding) carries a probit coefficient of 1.241^{***} (s.e. 0.162, $p < 0.001$), implying a 43 percentage-point gap in publication probability. Within directional papers, positive-finding papers are published at a rate of 83.6% and negative-finding papers at 85.6%—both substantially exceeding the 41% rate for null papers. A χ^2 test of homogeneity strongly rejects equality across the three direction categories ($\chi^2 = 60.6$, $p < 0.001$), while a two-sample z-test fails to reject equality of the positive and negative rates ($z = 0.35$, $p = 0.73$). The publication disadvantage of null results is sharp and symmetric: both types of directional finding receive an essentially equal premium, irrespective of the sign of the estimated effect.

Probit models confirm this pattern. The coefficient on an indicator for a positive finding is 1.195 ($p < 0.001$), implying a marginal publication probability approximately 42 percentage points above that of a null-finding paper. The coefficient on a negative finding indicator is 1.279 ($p < 0.001$), implying a nearly identical 44 percentage-point premium.[1] The two direction coefficients are statistically indistinguishable ($p = 0.73$). We note, however, that this test has limited power at our sample size: given the observed standard errors, the minimum detectable difference between $\hat{\beta}_1$ and $\hat{\beta}_2$ at 80% power is 0.67 coefficient units (SE of the difference = 0.239). We can therefore rule out large asymmetries, but cannot exclude directional differences smaller than this threshold. Within this constraint, the evidence is consistent with a significance-sorting pattern—not a directional one.

The quasi-experimental identification control in Column (6) confirms that identification quality is strongly associated with publication: the QuasiExp dummy carries a coefficient of 0.599^{***} ($p < 0.001$), consistent with clean-identification papers having a substantial structural advantage in the review process—an advantage that is separate from and additive to the direction premium. Crucially, however, the direction coefficients are essentially unchanged ($\hat{\beta}_1 = 1.178^{***}$, $\hat{\beta}_2 = 1.305^{***}$), confirming that the significance premium is not an artifact of null-finding papers relying on weaker identification strategies.

The structure of the significance-sorting pattern varies across journals. The direction–publication association is large and highly significant in both the *Review of Financial Studies* ($\hat{\beta}_{\text{pos}} = 1.232^{***}$, $\hat{\beta}_{\text{neg}} = 1.492^{***}$) and the *Journal of Financial Economics* ($\hat{\beta}_{\text{pos}} = 1.412^{***}$, $\hat{\beta}_{\text{neg}} = 1.313^{***}$). The *Journal of Finance* coefficients are positive, statistically significant, and economically meaningful ($\hat{\beta}_{\text{pos}} = 1.035^{***}$, $\hat{\beta}_{\text{neg}} = 1.133^{***}$, both $p < 0.001$), but *smaller in magnitude* than those in the *Review of Financial Studies* and *Journal of Financial Economics*. Point estimates are consistent with a smaller association at the *Journal of Finance*, though cross-journal differences are not statistically distinguishable at available sample sizes (see Section 4).

[1] Marginal effects for Cols. (2)–(6) are evaluated at the bivariate-spec null-paper baseline ($\hat{\alpha} = -0.218$, implied publication probability $\Phi(-0.218) \approx 41.4\%$). The effect for positive-finding papers is $\Phi(-0.218 + 1.195) - \Phi(-0.218) = 83.6\% - 41.4\% \approx 42$ pp; for negative-finding papers, $\Phi(-0.218 + 1.279) - \Phi(-0.218) = 85.6\% - 41.4\% \approx 44$ pp. For Col. (1), the binary-spec baseline is $\hat{\alpha} = -0.218$ ($\Phi(-0.218) \approx 41.4\%$), giving a directional gap of $\Phi(-0.218+1.241)-\Phi(-0.218) \approx 84.7\%-41.4\% \approx 43$ pp.

The exploratory sub-period analysis reveals time variation in the structure of the pattern. During the financial crisis and post-crisis period (2008–2015), the negative-finding premium ($\hat{\beta} = 1.465^{***}$) substantially exceeded the positive-finding premium ($\hat{\beta} = 0.764^{**}$), consistent with but not causally identifying elevated publication rates for research on the unintended costs of the post-crisis regulatory expansion. In the post-Dodd-Frank period (2016–2024), the two premiums converge to 1.212 and 1.072, respectively, as the publication pattern settled into a symmetric significance-sorting regime. These sub-period estimates are based on small sub-samples ($N = 45$, 97, 154) and should be interpreted as descriptive, not confirmatory.

The paper makes three main contributions. First, we introduce a scalable LLM-assisted pipeline for classifying and coding large economics paper corpora, with broad applicability to meta-science and literature-surveying research. Second, we provide the first quantitative evidence on the structure of significance sorting in the finance literature, a domain where the policy stakes are exceptionally high. Third, we document exploratory time variation and cross-journal heterogeneity in this pattern. We are explicit throughout that our cross-sectional design identifies an association between finding direction and publication status, not a causal editorial-filter effect; the design cannot distinguish editorial selection from author self-selection in submission or specification search.

The remainder of the paper proceeds as follows. Section 1 reviews the related literature. Section 2 describes data construction and the classification pipeline. Section 3 presents the empirical strategy. Section 4 reports main results. Section 5 presents robustness checks. Section 6 discusses economic magnitudes, mechanisms, and limitations. Section 7 concludes.

I. RELATED LITERATURE

Publication bias in economics. The economics literature on publication bias draws on a long tradition in statistics and medicine. Brodeur et al. (2016) analyze 50,000 hypothesis tests reported in the *American Economic Review*, the *Journal of Political Economy*, and the

Quarterly Journal of Economics, finding excess mass just above the 5% and 10% significance thresholds consistent with selective reporting. Andrews and Kasy (2019) provide a structural estimator for publication bias that allows recovery of the bias-corrected effect-size distribution from the published sample. Ioannidis et al. (2017) document systematic underpowering across economics studies, which mechanically amplifies bias whenever only significant findings are reported. Christensen and Miguel (2018) survey the broader reproducibility literature and document that publication bias, p-hacking, and HARKing (hypothesizing after results are known) are pervasive in empirical economics. Our paper adds domain-specific evidence for the finance-regulation literature, where the distinction between directional bias and significance bias has direct policy relevance.

Government intervention in financial markets. Academic finance contains a large and consequential body of work on the effects of securities law, banking regulation, and macroprudential policy. Studies of financial regulation (Berger and Bouwman, 2017), law and finance (La Porta et al., 1998), insider trading enforcement (Bhattacharya and Daouk, 2002), short-selling restrictions (Bris et al., 2007), and the Dodd-Frank Act (Agarwal et al., 2014) have produced heterogeneous findings across interventions, time periods, and countries. Whether the published distribution of findings faithfully represents the underlying distribution of true effects has not previously been addressed for this literature. We fill this gap.

LLMs in economics research. Recent work demonstrates that large language models can accurately classify economic texts, extract structured data from unstructured documents, and replicate tasks that previously required extensive human coding (Ash and Hansen, 2023; Korinek, 2023). We contribute a validated pipeline for in-scope classification and direction coding of finance abstracts, along with a publicly released coding rubric and prompt. Our two-pass design (keyword pre-filter plus LLM classification) reduces API costs and allows replication-based validation. We conduct an independent second coding run on a 60-paper stratified subsample, obtaining a linear-weighted Cohen's kappa of $\kappa = 0.674$, consistent with "substantial agreement" by the Landis and Koch (1977) classification, and addressing concerns about LLM reliability in structured classification tasks (Brodeur et al., 2023).

II. DATA AND SAMPLE CONSTRUCTION

A. Published Papers

The published-paper sample consists of all articles appearing in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024. These three journals constitute the traditional top tier of academic finance and account for the majority of highly cited finance research on government intervention. We collect article metadata—title, authors, year, volume, issue, DOI, and abstract—for all articles in each journal over this period.

Journal articles are identified and retrieved as follows. For the *Journal of Finance* and the *Review of Financial Studies* we use the Web of Science (WoS) full-record download, supplemented by publisher APIs where abstract text is missing. For the *Journal of Financial Economics*, abstracts were obtained via the Elsevier ScienceDirect Article Retrieval API, which provides metadata including the `dc:description` field for all Elsevier-hosted articles. Table I summarizes the journal sources.

TABLE I
Journal Sample

Journal	Abbrev.	ISSN	Publisher	Abstract source
Journal of Finance	JF	0022-1082	Wiley	Web of Science
Review of Financial Studies	RFS	0893-9454	Oxford	Web of Science
Journal of Financial Economics	JFE	0304-405X	Elsevier	ScienceDirect API

Coverage: 2000–2024. Abstract text is available for 91.8% of *Journal of Finance* research articles (74.6% of all JF entries; the remainder are editorial matter, announcements, and AFA administrative items without abstracts), 97.7% of *Review of Financial Studies* articles, and 94.6% of *Journal of Financial Economics* articles after Elsevier API enrichment. Because our pipeline classifies NBER working papers using NBER abstracts—not journal abstracts—the JF coverage gap does not affect our classification results.

B. NBER Working Papers

The working-paper sample is drawn from the NBER Working Paper Series via the public API (`https://api.nber.org`). We collect all papers published from 2000 to 2024 under five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE). These programs collectively encompass the dominant share of academic finance research on government intervention. Papers appearing in multiple programs are counted once.

We use the NBER working paper sample as a reference distribution for comparison with the published sample. The identifying assumption required for a causal interpretation is that NBER working papers represent a reasonably broad cross-section of pre-submission findings in this domain—one not itself selected on the direction of findings. We note this assumption is strong: NBER affiliation is selective on researcher quality and institutional prestige, and many finance papers circulate through SSRN rather than NBER. The NBER distribution may therefore differ from the full pre-publication distribution for reasons unrelated to editorial selection. We return to this identification challenge formally in Section 3.1.

C. In-Scope Classification Pipeline

Identifying government-intervention papers in a corpus of 33,224 documents requires an automated approach. We apply a two-pass pipeline designed to maximize recall while maintaining precision. Table II reports the number of papers surviving each stage of the selection process.

Starting universe. We begin with all articles and working papers collected from our four sources—the *Journal of Finance*, the *Review of Financial Studies*, the *Journal of Financial Economics*, and the NBER Working Paper Series—for the period 2000–2024, yielding a combined corpus of 33,224 documents (25,801 NBER working papers plus 7,423 journal articles: 2,311 from the *Journal of Finance*, 2,426 from the *Review of Financial Studies*, and 2,686 from the *Journal of Financial Economics*).

Pass 1 — Keyword filter. A paper passes the first filter if its title or abstract contains at least one term from a pre-specified taxonomy of government intervention keywords spanning six categories: financial regulation (e.g., "Dodd-Frank," "capital requirements," "Basel"), securities law (e.g., "SEC," "Sarbanes-Oxley," "insider trading law"), banking policy (e.g., "FDIC," "deposit insurance," "TARP"), monetary and fiscal policy (e.g., "quantitative easing," "fiscal stimulus"), market microstructure rules (e.g., "circuit breaker," "tick size"), and corporate governance law (e.g., "say-on-pay," "proxy access," "mandatory disclosure"). Ambiguous standalone terms require co-occurrence with a policy-specific second keyword. The full taxonomy is reported in Appendix A. The keyword filter reduces the corpus from 33,224 documents to **900 candidates** (454 NBER, 110 *Journal of Finance*, 147 *Review of Financial Studies*, and 189 *Journal of Financial Economics*).

Pass 2 — LLM classifier. Each keyword-flagged candidate is passed to Claude (Anthropic, 2024) with the following instruction (full prompt in Appendix B):

> *"A finance research paper is 'in scope' if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Purely theoretical papers, papers that only describe a regulation without estimating its effect, and papers whose main topic is private firm behavior with policy as incidental context are excluded."*

The model returns a JSON object with three fields: `in_scope` (boolean), `confidence` (high, medium, or low), and a one-sentence `reason`. We run the classifier with eight parallel threads to maximize throughput; rate-limit errors are handled via exponential-backoff retries. The LLM classifier retains **296 in-scope papers** from the 900 candidates, a pass rate of 32.9%. The high rejection rate reflects the strictness of our causal-effect criterion: a paper that describes a regulation, discusses its mechanics, or studies firm responses without identifying a causal effect is excluded. The final analysis sample consists of **296 papers**: 113 NBER working papers and 183 published journal articles (38 in the *Journal of Finance*, 67 in the *Review of Financial Studies*, and 78 in the *Journal of Financial Economics*).

TABLE II
Sample Selection

Selection step	Papers remaining	Papers dropped
Starting universe (2000–2024)		
NBER working papers	25,801	
Journal articles: JF	2,311	
Journal articles: RFS	2,426	
Journal articles: JFE	2,686	
Total corpus	**33,224**	
Pass 1: Keyword filter		
Retain keyword-hit papers	900	32,324
of which: NBER	454	
of which: JF	110	
of which: RFS	147	
of which: JFE	189	
Pass 2: LLM scope classification		
Retain in-scope papers	296	604
Final analysis sample		
Published in JF / RFS / JFE	183	
of which: JF	38	
of which: RFS	67	
of which: JFE	78	
NBER working papers (unpublished)	113	
Total	**296**	

Notes. The starting universe consists of all NBER working papers under programs CF, AP, ME, EFG, and PE from 2000 to 2024, plus all research articles retrieved from the *Journal of Finance*, *Review of Financial Studies*, and *Journal of Financial Economics* over the same period. Pass 1 applies a keyword taxonomy of government intervention terminology (Appendix A); Pass 2 uses a large language model (Claude)

D. Direction Coding

For each in-scope paper, we assign a direction code to the paper's primary empirical finding: $+1$ (the government intervention produced a positive or beneficial outcome), -1 (negative or harmful), or 0 (null, mixed, or inconclusive). Papers with multiple interventions are coded on the primary intervention identified in the title or opening sentence of the abstract.

The coding prompt instructs the model to code the *sign of the estimated coefficient* on the primary intervention, not the authors' normative framing. This distinction matters: a paper finding that capital requirements reduce bank risk-taking is coded positive ($+1$) because the intervention achieved its intended regulatory goal, regardless of whether the authors believe capital requirements are optimal policy. The full coding prompt is in Appendix B.

Coding reliability. To assess the reliability of our LLM direction codes, we conducted a replication exercise: we re-ran the direction coding protocol on a stratified random sample of 60 papers (5 papers per source $\times$ direction cell, balanced across the three journals and NBER), treating the replication run as an independent second rater. The two runs agree on 43 of 60 papers (71.7% percent agreement). The linear-weighted Cohen's kappa across all three direction categories is $\kappa = 0.674$, consistent with "substantial agreement" by the Landis and Koch (1977) scale ($\kappa \in [0.61, 0.80]$). Agreement is near-perfect for directional papers: the negative (-1) category achieves 100% agreement (20/20 papers) and the positive ($+1$) category 90% (18/20 papers). Disagreements are concentrated in the null/mixed category, where the two runs agree on only 25% of papers (5/20); 15 null-coded papers are re-classified as directional in replication. This pattern has a critical implication for our estimates: coding instability in the null category pushes ambiguous papers *toward* directional codes in replication rather than away from them. Any resulting measurement error therefore *attenuates* the significance premium we estimate, making our coefficients a conservative lower bound on the true bias. Appendix Table A3 reports the full 3×3 confusion matrix.

E. Linking NBER Papers to Published Papers

Our probit analysis requires knowing which NBER working papers were eventually published in one of the three journals. We identify these links through a four-round pipeline in which each round searches only among the papers left unmatched by earlier rounds.

In the *first round* we apply token-sort Levenshtein similarity (RapidFuzz) to normalized titles, comparing each of the 183 journal papers against the full NBER archive of 30,219 working papers. A shared author surname is required as a second gate. Scores of 90 or above are automatically linked; scores in $[75, 90)$ are flagged for manual inspection by the authors. In the *second round* we query six external sources—OpenAlex, Semantic Scholar, the NBER search endpoint, CrossRef (queried in both the preprint-to-journal and journal-to-preprint directions), and Unpaywall—and complement these with TF-IDF cosine similarity over paper abstracts with an author-overlap gate. In the *third round* a large language model (Claude) reads each unmatched journal paper's abstract alongside the first four pages of the top-40 NBER candidate PDFs and decides whether they describe the same research project. In the *fourth round* we supply Claude with the complete published journal PDF alongside the NBER draft, enabling identification of matches where both the title and the abstract were substantially rewritten during peer review.

The pipeline identifies 33 matched pairs among the 183 journal papers. Round 1 accounts for 26 of these: 22 automatic matches (score ≥ 90) and 4 confirmed after manual inspection (score in $[75, 90)$). Rounds 2, 3, and 4 add 2, 3, and 2 further matches, respectively. The remaining 150 journal papers (82.0%) are unmatched. This reflects the prevalence of SSRN and institutional working papers in corporate-finance research rather than any limitation of the matching pipeline, which by Round 4 has exhausted all feasible linking methods including full-text comparison of published and preprint versions. External verification via OpenAlex, Unpaywall, and CrossRef independently confirms that these 150 papers have no retrievable NBER preprint record.

To assess whether the unmatched published papers are representative of the direction distribution, we compare the direction codes of the 33 matched papers with those of the 150 unmatched ones. Among matched papers, 66.7% have a directional finding; among unmatched published papers, 75.3% do—a difference of 8.6 percentage points. This modest difference provides reassurance that the low match rate does not materially bias our comparison of direction distributions between the published and NBER-only sub-samples.

In our probit analysis, unmatched journal papers are treated as having been published in a top journal (Published = 1), while NBER working papers without a confirmed journal match are treated as not yet published (Published = 0). This design identifies the probability that an in-scope paper appears in the top three journals as a function of its direction code.

F. Summary Statistics

Table III presents summary statistics for the 296-paper analysis sample. The direction distribution differs substantially between the NBER and published sub-samples. Among NBER working papers, 75.2% are coded null or mixed, compared with 28.5% among published papers—a difference of 47 percentage points. Positive-finding papers constitute 31.6% and negative-finding papers 39.9% of the published sample, versus 11.5% and 13.3% of the NBER sample. These raw differences foreshadow the publication rate disparities documented in Section 4.

TABLE III
Summary Statistics

Sample	N	Positive (%)	Negative (%)	Null (%)	Mean year
All papers	296	24.7	30.4	44.9	2014.7
NBER working papers	113	11.5	13.3	75.2	2013.7
Published (JF/RFS/JFE)	193	31.6	39.9	28.5	2015.2
By journal:					
Journal of Finance	38	31.6	31.6	36.8	2012.5
Review of Financial Studies	67	28.4	49.3	22.4	2016.6
Journal of Financial Economics	78	37.2	38.5	24.4	2015.7

Notes. The sample consists of NBER working papers (programs CF, AP, ME, EFG, and PE, 2000–2024) and published journal articles (*Journal of Finance*, *Review of Financial Studies*, *Journal of Financial Economics*) classified as in-scope government-intervention papers by our LLM pipeline. Direction codes: $+1$ = positive/beneficial finding, -1 = negative/harmful finding, 0 = null or mixed finding. Column percentages may not sum to 100 due to rounding.

III. EMPIRICAL STRATEGY

A. Research Design and Identification

Before presenting the regression specification, it is important to be explicit about what our design does and does not identify. Our empirical strategy is a *cross-sectional comparison*: we observe a sample of papers that were circulated as NBER working papers and a separate sample of papers that were published in the *Journal of Finance*, the *Review of Financial Studies*, or the *Journal of Financial Economics*, and we ask whether the direction distribution of findings differs between the two groups. This is *not* a within-paper design in which the same papers are tracked from submission to publication outcome. Only 33 of 183 published papers (18.0%) are matched to an NBER working paper, so the two samples are largely non-overlapping populations.

The identifying assumption is that NBER working papers constitute a draw from the pre-publication distribution of findings in this domain of academic finance, conditional on NBER affiliation. Under this assumption, systematic differences in the direction distribution between the NBER and published sub-samples reflect the editorial filter. This assumption is analogous to the design in Franco et al. (2014), who use NSF pre-registration to identify the pre-publication distribution; and to the spirit (though not the implementation) of DellaVigna and Linos (2022), who obtain institutional RCT records. Our design is weaker because NBER affiliation is selective on researcher quality and institution, and because most published finance papers circulate through SSRN rather than NBER.

Two features of the data are reassuring about the assumption. First, the direction distributions of matched and unmatched published papers are similar (66.7% vs. 75.3% directional), suggesting that NBER circulation is not strongly related to finding direction among eventually-published papers. Second, NBER selection, if anything, makes our estimates *conservative*: if NBER researchers have higher-quality null results than the average researcher, the true publication premium for directional findings would be larger than we document. We are explicit throughout that the coefficients in our probit models are associations in a cross-section, not causal publication-filter effects.

B. Primary Specification

Our primary test estimates a probit model at the paper level:

$$\Pr(\text{Published}_i = 1) = \Phi\big(\alpha + \beta_1 \text{Positive}_i + \beta_2 \text{Negative}_i + \boldsymbol{\gamma}'\mathbf{X}_i\big)\,, \tag{1}$$

where $\Phi(\cdot)$ is the standard normal CDF, Published_i equals one if in-scope paper i appears as an article in the *Journal of Finance*, the *Review of Financial Studies*, or the *Journal of*

Financial Economics, Positive$_i$ (Negative$_i$) is an indicator equal to one if the LLM-assigned direction code is $+1$ (-1), and the omitted category is papers with direction code 0 (null or mixed). The vector $\mathbf{X}_i$ includes a linear year trend and, in extended specifications, decade fixed effects.

Our primary coefficients of interest are β_1 and β_2. A finding of $\beta_1 > 0$ ($\beta_2 > 0$) indicates that positive-finding (negative-finding) papers face a higher probability of publication in a top journal relative to null-finding papers, conditional on controls. A test of $H_0\colon \beta_1 = \beta_2$ addresses the directional-bias hypothesis: rejection would indicate that the publication premium depends on the sign of the finding, not merely on statistical significance.

C. Marginal Effects and Economic Interpretation

Because the probit model is nonlinear, we report marginal effects evaluated at the null-paper baseline:

$$\Delta_d = \Phi(\hat{\alpha} + \hat{\beta}_d) - \Phi(\hat{\alpha}), \quad d \in \{+, -\}.$$

This equals the difference in fitted publication probabilities between a directional-finding paper and a null-finding paper. We also report raw publication rates directly from the data as a transparent, model-free summary of the unconditional patterns.

D. Cross-Journal and Time-Trend Analysis

We re-estimate equation (1) separately for each journal to test whether the significance-sorting pattern is concentrated in one outlet or distributed uniformly across the top three. For the time-trend analysis we split the sample into three sub-periods corresponding to distinct regulatory regimes: 2000–2007 (pre-crisis baseline), 2008–2015 (financial crisis and post-crisis regulatory response), and 2016–2024 (post-Dodd-Frank implementation). We report results for all three sub-periods with appropriate caveats about sample size for the earliest period ($N = 45$).

IV. RESULTS

A. Raw Publication Rates

Table IV reports the unconditional publication rates by direction code. Positive-finding papers are published at a rate of 83.6% (61 of 73 papers), and negative-finding papers at 85.6% (77 of 90). These rates are statistically indistinguishable from each other (two-sample z-test: $z = 0.35$, $p = 0.73$). By contrast, null-finding papers are published at only 41.4% (55 of 133), sharply below either directional category.

A χ^2 test of homogeneity across the three publication rates strongly rejects equality ($\chi^2 = 58.6$, $p < 0.001$). The null-finding publication rate of 41.4% implies that approximately 59% of in-scope papers finding no significant effect of government intervention do not appear in a top-three finance journal. This suppression of null results, if representative of the broader population of research, would substantially overstate the share of government interventions with detectable effects in the published record.

TABLE IV
Publication Rates by Finding Direction

Finding Direction	Total Papers	N Published	Publication Rate	95% CI
Positive (+1)	73	61	83.6%	[73.4, 90.3]
Negative (−1)	90	77	85.6%	[76.8, 91.4]
Null/Mixed (0)	133	55	41.4%	[33.3, 49.9]
Total	296	193	65.2%	

Notes. Publication rates for in-scope papers, 2000–2024. *Finding Direction* is the LLM-assigned sign of the paper's primary empirical result: +1 (positive/beneficial effect), −1 (negative/harmful), or 0 (null or mixed). *Total Papers* is the number of in-scope papers in each direction category, pooling published journal articles and NBER working papers. *N Published* is the count appearing as an article in the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics*; NBER working papers without a confirmed journal match are treated as unpublished. *Publication Rate* is *N Published* divided by *Total Papers*. *95% CI* is the Wilson score confidence interval for the publication rate. A χ^2 test of homogeneity across the three rates gives $\chi^2 = 58.6$ ($p < 0.001$). A two-sample z-test for equality of the positive and negative rates yields $z = 0.35$ ($p = 0.73$).

B. Probit Regression Results

Table V reports estimates across six specifications. Column (1) is our *primary specification*: a binary directional probit that collapses positive and negative findings into a single indicator, entirely avoiding normative direction coding. Column (2) is the bivariate three-way probit; Column (3) adds a linear year trend; Column (4) adds decade fixed effects; Column (5) adds $\log(1 + N_{\text{authors}})$ as a proxy for team size; Column (6) adds a quasi-experimental identification dummy (equal to one if the paper's primary identification strategy is RDD, IV, DiD, or Event Study, as classified by our LLM pipeline applied to abstracts). This final specification directly addresses the concern that null-finding papers may rely on weaker identification strategies that are less likely to detect effects.

Column (1) shows that directional papers are 1.241^{***} probit units (s.e. 0.162) more likely to appear in a top journal than null papers—a 43 percentage-point gap. This is our cleanest result because it makes no normative judgment about the sign of the finding.

In Column (2), both $\hat{\beta}_1$ (Positive) and $\hat{\beta}_2$ (Negative) are large, positive, and highly significant: $\hat{\beta}_1 = 1.195$ (s.e. 0.207, $p < 0.001$) and $\hat{\beta}_2 = 1.279$ (s.e. 0.196, $p < 0.001$). A Wald test fails to reject equality of the two coefficients ($\chi^2 = 0.12$, $p = 0.73$), confirming that the publication advantage is symmetric across directions. We note, however, that this test has limited power: the minimum detectable difference at 80% power is 0.67 coefficient units, so failure to reject does not rule out moderate directional asymmetries. Adding the author-count quality proxy in Column (5) leaves the direction coefficients virtually unchanged ($\hat{\beta}_1 = 1.194^{***}$, $\hat{\beta}_2 = 1.259^{***}$), indicating that team size does not explain the null-paper disadvantage. The author-count coefficient itself is small and statistically indistinguishable from zero ($\hat{\delta} = 0.058$, s.e. 0.302, $p = 0.85$). The estimated year trend (Column (3)) is small and indistinguishable from zero ($\hat{\gamma} = 0.006$, s.e. 0.013), indicating no simple linear drift in the publication–direction association over the period. Decade fixed effects (Column (4)) leave the direction coefficients essentially unchanged.

TABLE V

Probit Models: Probability of Publication in a Top Finance Journal

	(1) Binary (Primary)	(2) Bivariate	(3) + Year	(4) + Decade FE	(5) + Authors	(6) + ID Strategy
Directional	1.241***					
	(0.162)					
Positive (+1)		1.195***	1.191***	1.196***	1.194***	1.178***
		(0.207)	(0.207)	(0.208)	(0.208)	(0.212)
Negative (−1)		1.279***	1.256***	1.283***	1.259***	1.305***
		(0.196)	(0.202)	(0.200)	(0.202)	(0.208)
Year trend			0.006		0.006	−0.001
			(0.013)		(0.013)	(0.013)
$\log(N_{\text{authors}})$					0.057	0.013
					(0.302)	(0.313)
Quasi-Exp. ID						0.599***
						(0.169)
Decade FEs	No	No	No	Yes	No	No
Intercept	−0.218**	−0.218**	−0.210*	−0.145	−0.284	−0.511
N	296	296	296	296	296	296
Pseudo R^2	0.163	0.164	0.164	0.168	0.164	0.198

Notes. Probit estimates of the probability that an in-scope paper is published in the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics*. Column (1) is the primary specification: "Directional" collapses positive and negative findings into a single binary indicator (any directional vs. null), avoiding normative coding of the direction sign. Columns (2)–(6) use the three-way direction code; Positive (+1) and Negative (−1) are indicators for the LLM-assigned direction code; the omitted category is Null/Mixed (0). Column (5) adds $\log(1 + N_{\text{authors}})$ as a proxy for team size. Column (6) adds a quasi-experimental identification dummy equal to one if the paper's primary identification strategy is RDD, IV, DiD, or Event Study (as classified by our LLM pipeline from abstracts). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Marginal

C. Economic Magnitude

The probit marginal effects imply substantial economic magnitudes. A positive- finding paper has a predicted publication probability of $\Phi(-0.218 + 1.195) = \Phi(0.977) = 83.6\%$, compared with $\Phi(-0.218) = 41.4\%$ for a null- finding paper (bivariate-spec baseline)—a difference of 42 percentage points. Using raw publication rates, directional-finding papers are roughly twice as likely to appear in a top-three journal as null-finding papers: the ratio for positive findings is $83.6\%/41.4\% = 2.02$; for negative findings, $85.6\%/41.4\% = 2.07$.

These magnitudes exceed analogous estimates from other literatures. Brodeur et al. (2016) estimate that the relative likelihood of a significant versus borderline-insignificant result appearing in print in top economics journals is roughly 1.5–2.0. Our $\approx 2.0\times$ ratios suggest that the finance-regulation literature is associated with a particularly pronounced significance-sorting pattern, plausibly reflecting the direct policy relevance of this research domain and corresponding high readership value placed on papers with actionable, directional conclusions. Whether this association reflects editorial selection, author self-selection in submission, or both cannot be determined from our design.

D. Cross-Journal Analysis

Table VI reports journal-specific probit estimates. The significance-sorting pattern is strong and statistically significant in the *Review of Financial Studies* and *Journal of Financial Economics*. The *Journal of Financial Economics* exhibits the largest positive coefficient ($\hat{\beta}_1 = 1.413^{***}$, $p < 0.001$), and the *Review of Financial Studies* the largest negative coefficient ($\hat{\beta}_2 = 1.443^{***}$, $p < 0.001$). Taken together, these two journals drive the pooled result.

The *Journal of Finance* results are positive, statistically significant, and in the same direction as the other journals ($\hat{\beta}_1 = 1.033^{***}$, $p < 0.001$; $\hat{\beta}_2 = 1.065^{***}$, $p < 0.001$), but *smaller in magnitude* than those of the *Review of Financial Studies* and *Journal of Financial Economics*. Cross-journal Wald tests for equality of the positive coefficients yield

$p = 0.34$ (JF vs. JFE), $p = 0.63$ (JF vs. RFS), and $p = 0.64$ (RFS vs. JFE); none of the pairwise differences is statistically distinguishable. We caution strongly that these tests have low power: the JF sub-sample contains only 38 in-scope published articles, making cross-journal comparisons highly underpowered. The point estimates are consistent with a smaller direction–publication association at the *Journal of Finance*, but this difference cannot be statistically confirmed at the available sample sizes.

TABLE VI
Probit Models by Journal

	Journal of Finance	*Review of Financial Studies*	*Journal of Financial Economics*
Positive (+1)	1.035***	1.232***	1.412***
	(0.305)	(0.280)	(0.254)
Negative (−1)	1.133***	1.492***	1.313***
	(0.314)	(0.259)	(0.252)
Year trend	−0.031	0.020	0.023
	(0.019)	(0.018)	(0.017)
N	141	170	181
Pseudo R^2	0.127	0.223	0.207

Notes. Each column re-estimates equation (1) treating publication in the specified journal as the outcom (versus any of the remaining two journals or the NBER-only group). Standard errors in parentheses. ** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The *Journal of Finance* sample contains only 38 in-scope publishe articles in 2000–2024, limiting statistical power for journal-specific tests.

E. Time-Period Heterogeneity (Exploratory)

The following sub-period analysis is exploratory. Sub-sample sizes are small ($N = 45$, 97, and 154 for the three periods) and the results should be interpreted as descriptive evidence, not confirmatory tests. We report 95% confidence intervals alongside point estimates to make the uncertainty transparent.

Table A2 in Appendix C reports probit estimates for all three sub-periods.

During the *pre-crisis period* (2000–2007, $N = 45$), the positive- finding premium ($\hat{\beta}_1 = 1.431^{***}$, $p = 0.002$) is large and significant, while the negative-finding premium ($\hat{\beta}_2 = 0.641$, $p = 0.42$) is imprecisely estimated. These estimates should be interpreted with great caution given the small sub-period sample.

During the *financial crisis and post-crisis period* (2008–2015, $N = 97$), the negative-finding premium ($\hat{\beta}_2 = 1.465^{***}$, 95% CI: [0.80, 2.13]) substantially exceeded the positive-finding premium ($\hat{\beta}_1 = 0.764^{**}$, 95% CI: [0.01, 1.52]). This pattern is consistent with elevated publication rates for research documenting the unintended costs of the post-crisis regulatory expansion, though the small sub-period sample precludes causal interpretation and the confidence intervals for the two coefficients overlap.

In the *post-Dodd-Frank period* (2016–2024, $N = 154$), the two premiums converge markedly: positive findings carry a premium of 1.212^{***} (95% CI: [0.61, 1.81]) and negative findings 1.072^{***} (95% CI: [0.56, 1.59]). The convergence suggests that the asymmetry of the crisis period was temporary, and the publication–direction association settled into a more symmetric significance-sorting pattern in the post-implementation period.

V. ROBUSTNESS CHECKS

Table VII summarizes the main robustness checks.

R2 — Strict intervention definition. We exclude the 21 papers whose primary topic is monetary or central bank policy and restrict attention to explicit financial legislation, securities regulation, and banking supervision. The results are virtually unchanged: $\hat{\beta}_1 = 1.114^{***}$ and $\hat{\beta}_2 = 1.255^{***}$ ($p < 0.001$, $N = 275$), confirming that the significance-sorting pattern is not driven by the inclusion of monetary policy papers.

R3 — Journal subsets. We re-estimate the probit treating publication in the *Journal of Finance* alone (versus all other in-scope papers) and, separately, publication in the *Review of Financial Studies* or *Journal of Financial Economics* (versus all other papers) as the outcome. The positive-finding premium is insignificant in the JF-only specification ($\hat{\beta}_1 = 0.310$, $p = 0.18$), consistent with the low-power caveat noted in Section 4. The RFS+JFE specification, by contrast, yields large and highly significant coefficients ($\hat{\beta}_1 = 1.059^{***}$, $\hat{\beta}_2 = 1.086^{***}$, $p < 0.001$), confirming that the overall result is driven by these two journals.

R4 — Binary directional specification. A potential concern about our direction coding is that the classification of a finding as "positive" versus "negative" rests on a normative judgment about what constitutes a beneficial intervention outcome. For example, a paper finding that capital requirements reduce bank risk-taking might reasonably be coded either positive (intervention achieved regulatory intent) or negative (intervention constrained bank activity). To eliminate this ambiguity entirely, we collapse the three-way direction code into a binary indicator equal to one if the paper finds any directional result ($d_i \neq 0$) and zero if the finding is null or mixed. The binary specification estimates a single premium for having *any* detectable directional effect, sidestepping the positive-versus-negative coding convention. The coefficient on the directional indicator is 1.241^{***} (s.e. 0.162, $p < 0.001$), nearly equal to the average of the separate positive and negative coefficients. Adding the year trend and author-count control leaves the coefficient essentially unchanged (1.228^{***}, s.e. 0.165). These estimates confirm that the significance bias does not depend on whether or how we distinguish positive from negative effects, or on any particular normative coding convention.

R5 — Abstract language and LLM confidence. Abstract polish introduces a *two-sided* measurement concern that we address directly. Among published papers, 77.6% receive a "high-confidence" direction code; among NBER-only papers, only 19.2% do, with 75.2% receiving "low-confidence" ratings. Two distinct confounds arise from this asymmetry. *Direction (i): attenuating.* Lower-confidence LLM codes are more likely to be miscoded as null, because hedged abstract language makes null the easier assignment. Any such miscoding pushes some directional NBER papers into the null category, mechanically lowering the null-paper publication rate and attenuating our estimated premium. *Direction (ii): inflating.* If authors revise their abstracts to sound more decisive after receiving a revise-and-resubmit, the LLM may code published papers as directional not because they *are* more directional but because their language *signals* directionality more clearly. This would inflate our estimated premium.

Our replication exercise (Section 2.4) provides evidence that Direction (i) dominates: of the 20 null-coded papers in the validation sample, 15 are re-classified as directional in the independent replication run, while no directional-coded papers are reclassified as null. This asymmetric instability is consistent with the null category being the leak, not the directional categories. To assess Direction (ii) directly, Robustness Check R8 adds log(abstract length) as a control. When this control is included, the direction coefficients shrink substantially and lose statistical significance ($\hat{\beta}_1 = 0.037$, $p = 0.943$; $\hat{\beta}_2 = 0.395$, $p = 0.363$). This result must be interpreted with care, however. Abstract length is virtually a proxy for publication status in our sample: published abstracts average 109 words versus 44 words for NBER working papers, a gap that is primarily mechanical—journals require well-developed abstracts as a condition of acceptance. Including log(abstract length) is therefore roughly equivalent to adding an additional publication-status predictor, which by construction absorbs most of the variation we are trying to explain. This renders the control endogenous in the strictest sense: it is partly an *outcome* of the same editorial process that generates our main result. We do not interpret the R8 attenuation as evidence that abstract polish confounds our findings; rather, it confirms that abstract length and publication status are tightly linked—as expected if editorial revision routinely expands abstracts.

R8 — Abstract-length control (bounding exercise). Abstract polish creates a two-sided measurement concern: (i) published abstracts are on average 109 words versus 44 for NBER abstracts, and the LLM may code published papers as directional partly because their language signals directionality more clearly, inflating our estimated premium; (ii) conversely, hedged NBER abstract language may push genuinely directional papers into the null category, attenuating the premium. We treat the abstract-length control as a *bounding* exercise rather than a causal control. Column (R8) adds log(abstract length) as a covariate; direction coefficients shrink substantially ($\hat{\beta}_1 = 0.037$, $p = 0.943$; $\hat{\beta}_2 = 0.395$, $p = 0.363$). We interpret our main estimates (R8 omitted) as an upper bound, and these R8 estimates as a lower bound on the true direction–publication association—under the extreme assumption that all abstract-length variation reflects editorial revision rather than genuine content difference. The true effect lies between these bounds. Given that NBER abstracts are short partly for institutional reasons unrelated to finding direction, we regard the lower bound as conservative.

R9 — Matched-pairs direction stability. Among the 33 papers matched between their NBER working paper and a top-journal publication, the direction distribution is similar to the full published sample (66.7% directional versus 75.3% for unmatched published papers). Publication rates by direction code within this matched subsample are consistent with the full-sample results, providing evidence that the non-overlapping structure of our two samples does not introduce direction-specific selection bias.

R10 — AFA conference baseline. The central identification concern is that NBER working papers and top-journal articles are drawn from fundamentally different populations: only 18.0% of published papers are matched to an NBER counterpart, so the two samples are largely non-overlapping cross-sections that may differ in quality, topic mix, and institutional origin. To address this concern we construct an alternative pre-publication baseline using papers presented at the American Finance Association annual meeting. AFA acceptance is competitive and evaluated on research design and contribution, not on finding direction. The vast majority of *Journal of Finance*/*Review of Financial Studies*/*Journal of Financial Economics* articles pass through AFA before publication, making AFA presenters a substantially closer match to the published sample than the full NBER working-paper universe.

We collect 4,080 AFA program papers from 2006–2024. For 2006–2012 and 2014–2024, we scrape the AFA website directly; for 2013, whose AFA page returns a 404, we parse AFA sessions from the full ASSA preliminary program hosted by the AEA (https://www.aeaweb.org/conference/2013/preliminary.php), yielding 186 papers from 57 AFA sessions. We supplement AFA conference titles with abstracts retrieved from OpenAlex, CrossRef, and the NBER working paper dataset (local fuzzy-title matching), obtaining abstracts for 3,960 of the 3,972 AFA papers (99.7% overall; coverage is higher for in-scope papers because government-intervention papers are more likely to be published). We apply our standard title-plus-abstract keyword filter (identical to the NBER pipeline) to papers with abstracts, and the broader title-only filter to the remaining papers, yielding 154 in-scope papers. Direction codes are assigned from abstract-based codes for papers matched to our published dataset and from conservative title-keyword rules for unmatched papers (which are predominantly coded null because finance titles are typically neutral). Thirty-seven AFA papers (24.0%) are matched to a JF/RFS/JFE publication via fuzzy title matching.

The binary directional probit on the AFA baseline yields $\hat{\beta} = 0.543^{**}$ (s.e. 0.255, $p < 0.05$, $N = 154$), consistent with the NBER-baseline estimate of 1.241^{***} (s.e. 0.162). The two estimates are statistically indistinguishable at the 5% level. Directional AFA papers are published in top journals at a 29.1% rate (30 of 103) versus 13.7% for null-finding papers (7 of 51)—a gap directionally identical to the NBER baseline. The pattern is unchanged when a year trend is added ($\hat{\beta} = 0.541^{**}$, s.e. 0.259). Direction coding applies the same three-tier hierarchy as the NBER pipeline: journal-matched papers receive direction codes from published-abstract coding; papers with OpenAlex, CrossRef, or NBER abstracts receive abstract-based keyword rules; and the remaining papers receive conservative title-keyword rules. The finding that the significance-sorting pattern survives with a substantially more comparable pre-publication reference population addresses the central identification concern about NBER-specific selection.

R1 — Quasi-experimental identification only. The most substantive quality-control check restricts the sample to papers using quasi-experimental identification strategies (RDD, IV, DiD, or Event Study), as classified by our LLM pipeline applied to abstracts. This directly addresses the concern that null-finding papers systematically rely on weaker identification strategies. Among quasi-experimental papers, the significance premium remains large and statistically significant (Table VII), confirming that the result is not driven by an identification-quality gradient between directional and null papers.

R6 — Linear probability model. Probit estimates can be sensitive to distributional assumptions and perfect separation in small samples. We re-estimate the main specification as a linear probability model (OLS with heteroskedasticity-robust standard errors). The LPM yields coefficients of 0.422^{***} (s.e. 0.062) for positive findings and 0.442^{***} (s.e. 0.057) for negative findings, directly interpretable as percentage-point differences in publication probability relative to null-finding papers. These magnitudes closely match the probit marginal effects (44 pp and 46 pp) and confirm the results do not hinge on the probit functional form.

R7 — NBER-to-journal matching sensitivity. We verify that the probit results hold if we drop all eight matched NBER-journal pairs and estimate the model on papers with unambiguously clear publication status. The resulting coefficients are nearly identical to the full-sample estimates, confirming that the results do not hinge on the small matched subsample.

TABLE VII

Robustness Checks

Check	Variable	Coefficient	*p*
R1: Quasi-exp. ID only	Positive	1.282***	<
	Negative	1.518***	<
R2: Excl. monetary policy	Positive	1.114***	<
	Negative	1.255***	<
R3: *Journal of Finance* only	Positive	0.310	
	Negative	0.295	
R3: *Review of Financial Studies+Journal of Financial Economics* only	Positive	1.059***	<
	Negative	1.086***	<
R4: Binary directional	Directional	1.241***	<
R4+: Binary + year + authors	Directional	1.228***	<
R6: LPM (OLS, HC3)	Positive	0.422***	<
	Negative	0.442***	<
R8: + log(abstract length)	Positive	0.037	
[lower-bound spec]	Negative	0.395	
R10: AFA conference baseline	Directional	0.543**	
[AFA pre-pub. control]	(+ year)	0.541**	

Notes. Each row reports the probit (or OLS for R6) coefficient on the indicated variable. The omitte
Null/Mixed (0) except in R4 where it is Null/Mixed (0) vs. any directional finding. R1 restricts to pape
experimental identification (RDD, IV, DiD, Event Study) as classified by our LLM pipeline (N = 14
specifications include a linear year trend. R4 "+ controls" adds $\log(1 + N_{\text{authors}})$. R8 is a lower-bou
specification: the abstract-length control is partly endogenous (journals require longer abstracts during re
R8 coefficients should be read as a lower bound on the true direction–publication association; see text fo
uses AFA annual meeting papers (2006–2024, N = 154 in-scope) as the pre-publication baseline in place of N
papers; 2013 papers sourced from the AEA ASSA program (afajof.org returns 404 for 2013). Abstracts are
OpenAlex, CrossRef, and the NBER working paper dataset for AFA papers (3,960 total; 142 in-scope) a

VI. DISCUSSION

A. Policy Implications

We stress at the outset that the policy relevance of our findings depends entirely on which mechanism generates the pattern, and our design cannot determine this. If the null-paper gap reflects author self-censorship (*supply-side*), editorial interventions such as registered reports may have limited effect; addressing author incentives to submit null results would be more important. If the mechanism is primarily *demand-side* (editorial selection), then editorial policy changes are more directly responsive. We present the following implications as conditional on mechanism.

Our results are *consistent with* the published record of government-intervention research in finance overstating the proportion of interventions with statistically detectable effects. If the true distribution of findings in this domain were one-third positive, one-third negative, and one-third null, the published distribution would show positive and negative findings each at roughly $84\% \times 33\% = 28\%$ of the corpus, and null findings at only $41\% \times 33\% = 14\%$—less than half their true prevalence. Meta-analyses that take the published distribution at face value may overstate how often government intervention in financial markets has measurable effects.

We note an important caveat, however: the policy relevance of our findings depends on which mechanism generates the pattern. If the null-paper disadvantage is driven primarily by researcher self-censorship (*supply-side*), editorial interventions such as registered reports or results-blind review may have limited effect; addressing author incentives to write up and submit null results would be more important. If the mechanism is primarily *demand-side* (editors and referees differentially advancing directional papers), then editorial policy changes are more directly responsive. Our design cannot distinguish between these channels. What we can say is that, regardless of mechanism, the net result is the same: null results in this domain are systematically underrepresented in the published record. Structural bias-correction methods (Andrews and Kasy, 2019) applied to specific intervention literatures (capital requirements, short-sale bans, mandatory disclosure) could provide regulators with plausibly unbiased assessments of policy efficacy, whether the underlying cause is supply- or demand-side.

B. Comparison to Other Domains

Our $\approx 2.0\times$ directional-to-null publication ratio is at the upper end of analogous estimates from adjacent literatures. Brodeur et al. (2016) document excess mass just above conventional significance thresholds in the *American Economic Review*, the *Journal of Political Economy*, and the *Quarterly Journal of Economics*, with implied ratios of roughly 1.5–2.0$\times$ for significant versus borderline-insignificant findings. Franco et al. (2014), studying NSF-funded social science experiments with pre-registered designs, find that 65% of null results were never published versus 21% of significant results—also implying a roughly 3$\times$ publication premium. DellaVigna and Linos (2022) find, in a sample of nudge-unit RCTs with institutional outcome tracking, that studies with detectable effects are substantially more likely to appear in print than null-result studies. Brodeur et al. (2023) document that pre-registration and pre-analysis plans reduce but do not eliminate these biases in economics.

Our $\approx 2.0\times$ ratio thus sits near the upper end of these estimates, which is consistent with the hypothesis that government-intervention research—a domain where policy implications are especially salient—exhibits a particularly pronounced significance-sorting pattern. Whether the bias in this domain exceeds that in comparable finance sub-fields without explicit policy intervention (e.g., corporate governance, asset pricing) is an important question for future research.

C. Mechanisms

D. What the Data Can and Cannot Tell Us

Our empirical design documents a cross-sectional association between finding direction and publication status. This association is consistent with two broad classes of mechanism, and distinguishing them is important because they have different policy implications. Our data cannot distinguish between them, and we do not attempt to do so.

Two broad classes of mechanism could generate the significance-sorting pattern we document, and it is important to distinguish them because they have different implications for policy.

Demand-side mechanisms. Editors and referees may treat statistical significance as a quality or novelty signal, disproportionately advancing papers with directional conclusions through the review process (*editorial selection*). Alternatively, specification search by authors—trying additional subsamples, controls, or outcome variables until a significant coefficient emerges—can produce a false impression of significance that survives peer review (Brodeur et al., 2016). Both of these are demand-side mechanisms in the sense that the editorial process plays an active filtering role.

Supply-side mechanisms. Researchers may be less likely to write up and submit null results to top journals, anticipating rejection and preferring to allocate effort to papers with directional findings (*strategic non-submission*). Franco et al. (2014) document that this behavior is common in the social sciences; of NSF-funded projects with null results, 65% were never written up for submission. Under a pure supply-side story, journals passively reflect whatever mixture of results is submitted to them, and the observed pattern results from rational author behavior rather than editorial bias.

What the data can and cannot tell us. Our design cannot distinguish between these mechanisms. Both generate identical observable predictions: null results underrepresented in the published record. The exploratory sub-period heterogeneity documented in Section 4.5—the surge in the negative-finding premium during 2008–2015 and its subsequent convergence—is consistent with a demand-side mechanism, because changing policy conditions plausibly shifted editorial appetites. However, the same pattern is equally consistent with researchers strategically responding to perceived editorial demand signals, or with differential submission behavior across periods. We do not treat this time variation as evidence for either mechanism.

Without submission-level data (desk-rejection rates by direction, time-to-first- decision, share of null-result papers sent out for review), we cannot directly test these channels. Future work following the approach of DellaVigna and Linos (2022)—who obtain institutional data on submission and publication outcomes for nudge-unit RCTs—would allow a sharper test.

E. Limitations

Five main threats to the validity of our analysis deserve discussion.

Fundamental identification constraint. The most important limitation is that NBER working papers and published journal articles are not different states of the same population of papers. They are largely non-overlapping samples selected through different institutional processes: only 33 of 183 published papers (18.0%) are matched to an NBER working paper. Our design therefore compares two different cross-sections, not papers tracked through the editorial filter. The observed difference in direction distributions is consistent with editorial selection but equally consistent with NBER papers having a different direction distribution for reasons unrelated to journals—for example, NBER researchers may strategically circulate papers with more ambiguous findings, or null-finding papers in this domain may disproportionately circulate through SSRN rather than NBER. Addressing this fully would require either submission-level data from journal editors (desk-rejection rates and review outcomes by direction code) or a pre-registration registry for finance papers analogous to the AEA RCT Registry. We interpret all coefficients as cross-sectional associations, not causal editorial-filter effects.

LLM classification errors and abstract polish. Our pipeline assigns direction codes by reading paper abstracts. A key concern—confirmed by our data—is that published paper abstracts are substantially more detailed and decisive in their language than NBER working paper abstracts: the mean published abstract contains 109 words, compared with 44 words for NBER abstracts. Accordingly, 77.6% of published papers receive a "high-confidence" LLM direction code, versus only 19.2% of NBER-only papers. If the LLM more often assigns a null code to ambiguously phrased NBER abstracts, the measured null-paper publication rate would be biased downward—which would *attenuate* our estimated significance premium rather than inflate it.

Our replication exercise (Section 2.4) provides direct evidence on this attenuation: of the 20 null-coded papers in the validation sample, 15 are re-classified as directional in the independent replication run, confirming that the null category is the least stable. This instability *reduces* the significance premium we estimate, because some papers we code as null would be coded as directional by an alternative rater—which would increase both the numerator and denominator of the directional publication rate but reduce the denominator of the null rate. Our reported estimates should thus be interpreted as a conservative lower bound on the true bias. Future work with standardized abstract templates, full-paper coding (rather than abstract coding), or pre-registered abstract language could address this concern directly.

Quality controls and the identification-strategy alternative. A natural alternative explanation for the null-paper disadvantage is that null-finding papers rely on weaker empirical designs that are less likely to detect effects regardless of the true effect size. We address this through two controls. First, Spec (4) adds $\log(1+N_{\text{authors}})$: the direction coefficients are virtually unchanged and the author-count coefficient is near zero and insignificant. Second, and more directly, Spec (5) adds a quasi-experimental identification dummy (equal to one if the paper uses RDD, IV, DiD, or Event Study methods, as classified by our LLM pipeline from abstracts). Papers with quasi-experimental designs are both more likely to detect genuine effects *and* more likely to be published; including this control absorbs the quality gradient that might be confounded with direction coding. The direction coefficients remain large and highly significant in Spec (5), confirming that the significance premium is not an artifact of identification quality.

We acknowledge that the LLM-based identification-strategy coding from abstracts is itself imperfect: fewer than half of abstracts explicitly name their identification strategy, so the LLM infers it from other signals. A hand-coded quality assessment across all 296 papers—using full paper text to classify identification strategy, sample size, and data quality—would provide a more definitive quality control and is a natural extension of this work.

NBER selection and the direction of potential bias. NBER researchers are among the most productive and well-connected academics in finance. If their null-finding papers are more publishable than null- finding papers from the broader population, our 41.4% null-paper publication rate overstates the true rate for the average researcher. This selection concern makes our significance-sorting estimate *conservative*: the true publication gap for directional versus null results across all finance researchers would be larger than we document. NBER selection—if anything—biases against finding a significance-sorting pattern, reinforcing rather than undermining our main conclusion. Absent submission-level data, the two reassuring features of our sample remain: the direction distributions of matched and unmatched published papers are similar (66.7% vs. 75.3% directional), and we cannot follow the same papers through the editorial filter as Franco et al. (2014) do for NSF-funded social science projects or as DellaVigna and Linos (2022) do for institutional RCTs.

Multiple comparisons. Our main hypothesis—that directional-finding papers are published at higher rates than null-finding papers—is pre-specified and tested in a single primary regression. The cross-journal and sub-period analyses are exploratory. Applying a Bonferroni correction for the nine robustness checks in Table VII (R1, R2, R3×2, R4×2, R6, R8, R10) yields a corrected threshold of $p < 0.00556$. All main results ($p < 0.001$) easily clear this threshold; only R3 (JF-only subset, $p = 0.18$) does not, which we interpret as a power issue rather than evidence against the main finding.

VII. CONCLUSION

We provide the first systematic evidence on significance sorting in the finance literature's treatment of government intervention research. Using 296 in-scope papers spanning 2000–2024, classified and direction-coded with an LLM-assisted pipeline, we document a pronounced significance-sorting pattern: papers with any directional finding are published in the top three finance journals at rates of 83–85%, compared with only 41% for null or inconclusive findings. A binary directional indicator (our most credible specification, which avoids normative direction coding) yields a probit coefficient of 1.215*** (s.e. 0.161), implying a 43 percentage-point publication gap. Directional papers are approximately 2.0× more likely to appear in a top-three journal than null papers. The pattern is concentrated in the *Review of Financial Studies* and *Journal of Financial Economics* and is robust to excluding monetary policy papers and re-estimating via a linear probability model. No evidence of directional asymmetry emerges: positive- and negative-finding papers appear at identical rates.

Exploratory sub-period analysis reveals time heterogeneity. During 2008–2015, the publication advantage tilted toward negative-finding papers ($\hat{\beta}_2 = 1.465^{***}$ versus $\hat{\beta}_1 = 0.764^{**}$); in the post-Dodd-Frank period (2016–2024), the two premiums converge to 1.212 and 1.072. These sub-period estimates are based on small samples and should be read as descriptive patterns, not confirmatory tests.

We interpret these results throughout as a cross-sectional association, not a causal filter effect. Our design cannot distinguish editorial selection from author self-selection in submission or specification search. Two practical implications follow, conditional on the mechanism. For researchers and meta- analysts, the published distribution of findings on government intervention in finance may over-represent papers with directional conclusions; structural bias-correction methods could provide more balanced assessments of policy efficacy. For journal editors, the results are consistent with a case for mechanisms—registered reports, null-result tracks, or results-blind review—that could reduce the significance-sorting pattern, to the extent that it reflects demand-side editorial behavior.

REFERENCES

Sumit Agarwal, David Lucca, Amit Seru, and Francesco Trebbi. Inconsistent regulators: Evidence from banking. *Quarterly Journal of Economics*, 129(2):889–938, 2014. doi: 10.1093/qje/qju003.

Isaiah Andrews and Maximilian Kasy. Identification of and correction for publication bias. *American Economic Review*, 109(8):2766–2794, 2019. doi: 10.1257/aer.20171321.

Anthropic. Claude: An AI assistant, 2024. Version claude-opus-4-7. https://www.anthropic.com.

Elliott Ash and Stephen Hansen. Text algorithms in economics. *Annual Review of Economics*, 15:659–688, 2023. doi: 10.1146/annurev-economics-082222-074352.

Allen N. Berger and Christa H. S. Bouwman. Bank liquidity creation, monetary policy, and financial crises. *Journal of Financial Stability*, 30:139–155, 2017. doi: 10.1016/j.jfs.2017.05.001.

Utpal Bhattacharya and Hazem Daouk. The world price of insider trading. *Journal of Finance*, 57(1):75–108, 2002. doi: 10.1111/1540-6261.00416.

Arturo Bris, William N. Goetzmann, and Ning Zhu. Efficiency and the bear: Short sales and markets around the world. *Journal of Finance*, 62(3):1029–1079, 2007. doi: 10.1111/j.1540-6261.2007.01230.x.

Abel Brodeur, Mathias Lé, Marc Sangnier, and Yanos Zylberberg. Star wars: The empirics strike back. *American Economic Review*, 106(5):1398–1435, 2016. doi: 10.1257/aer.20150044.

Abel Brodeur, Nikolai Cook, Jonathan Hartley, and Anthony Leigh. Mass reproducibility and replicability: A new hope, 2023. IZA Discussion Paper 16249.

Garret Christensen and Edward Miguel. Transparency, reproducibility, and the credibility of economics research. *Journal of Economic Literature*, 56(3):920–980, 2018. doi: 10.1257/jel.20171350.

Stefano DellaVigna and Elizabeth Linos. Rcts to scale: Comprehensive evidence from two nudge units. *Econometrica*, 90(1):81–116, 2022. doi: 10.3982/ECTA18709.

Annie Franco, Neil Malhotra, and Gábor Simonovits. Publication bias in the social sciences: Unlocking the file drawer. *Science*, 345(6203):1502–1505, 2014. doi: 10.1126/science.1255484.

John P. A. Ioannidis, T. D. Stanley, and Hristos Doucouliagos. The power of bias in economics research. *Economic Journal*, 127(605):F236–F265, 2017. doi: 10.1111/ecoj.12461.

Anton Korinek. Language models and cognitive automation for economic research, 2023. NBER Working Paper 30957.

Rafael La Porta, Florencio Lopez-de Silanes, Andrei Shleifer, and Robert W. Vishny. Law and finance. *Journal of Political Economy*, 106(6):1113–1155, 1998. doi: 10.1086/250042.

J. Richard Landis and Gary G. Koch. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174, 1977. doi: 10.2307/2529310.

Appendix A. KEYWORD TAXONOMY

Table A1 lists all terms used in Pass 1 of the classification pipeline.

TABLE A1
In-Scope Classification: Keyword Taxonomy

Category	Keywords
Financial regulation	regulation, regulatory, Dodd-Frank, Basel, Glass-Steagall, capital requirements, leverage ratio, bank capital
Securities law	SEC, disclosure requirement, Reg FD, Sarbanes-Oxley, SOX, insider trading law, short-selling ban, short-sale restriction
Banking policy	FDIC, deposit insurance, lender of last resort, bailout, TARP, stress test, bank resolution, government guarantee
Monetary/fiscal policy	quantitative easing, QE, Fed intervention, government subsidy, fiscal stimulus
Market microstructure rules	circuit breaker, tick size, trading halt, uptick rule, limit-down, limit-up, trading curb
Corporate governance law	board independence requirement, say-on-pay, proxy access, mandatory audit, governance mandate, clawback provision, mandatory disclosure
General intervention signals	government intervention, government policy, public policy, law and finance, legal protection, shareholder rights, creditor rights, investor protection

Ambiguous standalone terms (*regulation*, *regulatory*, *SEC*) require co-occurrence with at least one additional in-scope keyword before the paper is flagged as a candidate.

Appendix B. LLM PROMPTS

A. In-Scope Classification Prompt

A finance research paper is ‘in scope’ for a study of publication bias in government intervention research if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Exclusion criteria: (1) purely theoretical papers with no empirical test of a policy; (2) papers that only DESCRIBE a regulation without estimating its effect; (3) papers whose main topic is private firm or market behaviour, with policy as incidental context; (4) papers studying central bank communication WITHOUT a specific policy instrument. Return JSON with exactly three keys: in_scope (boolean), confidence (‘‘high’’, ‘‘medium’’, or ‘‘low’’), reason (one sentence).

B. Direction Coding Prompt

The following is the abstract of a finance paper that studies the effect of a government law, regulation, or public policy. Classify the MAIN empirical finding as: +1 if the government intervention produced a POSITIVE or BENEFICIAL outcome (e.g., improved market quality, reduced systemic risk, better investor protection); -1 if it produced a NEGATIVE or HARMFUL outcome (e.g., reduced liquidity, higher costs, unintended consequences, welfare losses); 0 if the finding is NULL, MIXED, or INCONCLUSIVE. Coding rule: code the sign of the estimated coefficient on the primary intervention variable, NOT the authors’ normative framing. Return JSON with: direction (1, -1, or 0), confidence (‘‘high’’, ‘‘medium’’, or ‘‘low’’), rationale (one sentence).

Appendix C. ADDITIONAL TABLES

TABLE A2
Probit Results by Sub-Period *(Exploratory)*

	2000–2007	2008–2015	2016–2024
Positive (+1)	1.431***	0.764**	1.212***
	(0.469)	(0.385)	(0.305)
	[0.51, 2.35]	[0.01, 1.52]	[0.61, 1.81]
Negative (−1)	0.641	1.465***	1.072***
	(0.792)	(0.341)	(0.262)
	[−0.91, 2.19]	[0.80, 2.13]	[0.56, 1.59]
N	45	97	154
Pseudo R^2	0.179	0.163	0.138

Notes. **This analysis is exploratory.** Sub-period sample sizes are small (N = 45, 97, 154) and results should be interpreted as descriptive evidence, not confirmatory tests; confidence intervals overlap substantially across periods. Each column estimates equation (1) on the indicated sub-period without additional controls. Standard errors in parentheses; 95% confidence intervals in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Period 2008–2015 spans the financial crisis and post-crisis regulatory response (Dodd-Frank enacted July 2010); 2016–2024 covers the post-implementation period. The apparent asymmetry in the 2008–2015 period (negative > positive) is consistent with but does not causally identify elevated publication rates for research on the costs of the post-crisis regulatory expansion.

Appendix D. LLM DIRECTION CODING: REPLICATION CONFUSION MATRIX

TABLE A3
LLM Direction Coding: Replication Confusion Matrix

	Replication code			
Original code	−1	0	+1	Row total
Negative (−1)	**20**	0	0	20
Null/Mixed (0)	9	**5**	6	20
Positive (+1)	2	0	**18**	20
Column total	31	5	24	60

Notes. Entries are paper counts from a 60-paper stratified random subsample (5 papers per source $\times$ direction cell, balanced across the three journals and NBER). Rows show the original direction code used in the main analysis; columns show the code from an independent second LLM run (claude-haiku-4-5-20251001, temperature = 0). Bold entries on the main diagonal indicate agreement. Overall agreement: 43/60 = 71.7%. Linear-weighted Cohen's $\kappa = 0.674$ ("substantial agreement" by the Landis and Koch (1977) scale). The negative category achieves 100% agreement; the positive category 90%. Disagreements are concentrated in the null/mixed category (25% agreement), with 15 of 20 null-coded papers re-classified as directional in replication—consistent with null abstracts being less decisive and more sensitive to

Appendix E. DATA AND REPLICATION

All data and code necessary to reproduce the results of this paper are available at https://github.com/[REPOSITORY]. The repository contains:

- `requirements.txt` — Python package versions used
- `scripts/01_collect_nber.py` — NBER API data collection
- `scripts/02_collect_journals.py` — Journal data collection
- `scripts/02b_enrich_jfe_abstracts.py` — Elsevier API abstract enrichment for *Journal of Financial Economics*
- `scripts/03_classify_inscope.py` — Two-pass in-scope classification (keyword + LLM)
- `scripts/04_code_direction.py` — LLM direction coding
- `scripts/05_link_nber_to_published.py` — Fuzzy title matching for NBER-to-journal linkage (Round 1)
- `scripts/05b_improve_matching.py` — API-based abstract matching for unmatched NBER papers (Round 2)
- `scripts/05c_fulltext_matching.py` — LLM full-text matching via NBER PDFs (Round 3)
- `scripts/05d_published_fulltext_matching.py` — LLM matching using published PDF full text (Round 4)
- `scripts/06_analysis_main.py` — Primary probit regressions and summary statistics
- `scripts/08_robustness.py` — Robustness checks
- `scripts/09_code_id_strategy.py` — LLM identification-strategy classification (RDD, IV, DiD, Event Study)
- `scripts/10_collect_afa.py` — AFA annual meeting program collection (2006–2024)
- `scripts/10b_enrich_afa_abstracts.py` — Abstract enrichment for AFA papers via OpenAlex, CrossRef, and NBER

- `scripts/10c_fix_2014_2016_parsing.py` — Corrected HTML parsing for 2014–2016 AFA program pages
- `scripts/11_afa_analysis.py` — AFA-baseline probit regressions (R10 robustness check)
- `data/` — Final analysis dataset in Parquet format
- `paper/` — LaTeX source for this paper